Package 'rexcelbridge'

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Type Package

Title Bridge Excel Add-in Formulas and RTD Feeds into R (Windows)

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Description Vendor-neutral helpers to evaluate Excel formulas (including RTD-based add-ins like DTN ProphetX and Bloomberg) from R via COM automation. Supports single-cell results with
a pluggable readiness predicate to handle transient ``Wait''
tokens from providers.

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URL https://github.com/you/rexcelbridge

BugReports https://github.com/you/rexcelbridge/issues

Encoding UTF-8

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Depends R (>= 4.1.0)

Imports RDCOMClient

Remotes github::omegahat/RDCOMClient

Suggests tidyr, mockery, knitr, rmarkdown, testthat (>= 3.0.0)

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Contents

rb_ensure_addin 2
rb_eval_single 2
rb_eval_with_retries 3
rb_excel_quote 5
rb_kill_excel 5
rb_ready_predicate 6
rb_start_excel 6
rexcelbridge_controls 7

Index **8**

`rb_ensure_addin` *Ensure an Excel add-in is loaded (best effort)*

Description

Tries to locate and enable an add-in by name pattern (case-insensitive). Useful before evaluating formulas that depend on a vendor add-in.

Usage

```
rb_ensure_addin(xl, pattern)
```

Arguments

`xl`	Excel Application COM handle from `rb_start_excel()`.
`pattern`	Character regex used to match the add-in name (e.g., "ProphetX", "Bloomberg").

Details

Defensive against environments where `AddIns` is missing or inaccessible.

Value

Invisibly returns TRUE (never errors if not found).

See Also

Other excel bridge: `rb_eval_single()`, `rb_eval_with_retries()`, `rb_excel_quote()`, `rb_kill_excel()`, `rb_ready_predicate()`, `rb_start_excel()`

`rb_eval_single` *Evaluate one or more single-cell Excel formulas*

Description

Writes each formula into successive rows of column A in a new workbook, polls until values are "ready" per a predicate, and returns the results.

Usage

```
rb_eval_single(
  formulas,
  timeout_sec = 1,
  visible = FALSE,
  ready_fn = rb_ready_predicate()
)
```

Arguments

formulas	Character vector of Excel formulas (e.g., "=SUM(1,2)", '=RTD("prophetx.rtdserver","","AIHIST",...) '=BDP("IBM US Equity","PX_LAST")').
timeout_sec	Max seconds to wait. Default 1.
visible	Show the Excel window. Default FALSE.
ready_fn	A readiness predicate built by rb_ready_predicate(). Default uses c("Wait","Loading...","N/A") as transient tokens.

Value

Data frame with columns: row, formula, result, ready.

See Also

Other excel bridge: rb_ensure_addin(), rb_eval_with_retries(), rb_excel_quote(), rb_kill_excel(), rb_ready_predicate(), rb_start_excel()

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rb_eval_with_retries *Evaluate Excel/ProphetX RTD queries with retries*

Description

Efficiently evaluates a vector of Excel or ProphetX RTD-style queries with retry logic, partial re-evaluation of failed queries, and optional parallel fallback for single-query execution. This function is used to safely handle transient Excel COM errors or timeouts when running multiple RTD calls through the bridge.

Key optimizations include:

- Deduplication of identical queries (evaluated once).
- Retrying only failed entries in subsequent attempts.
- Conditional restarting of Excel only after failed attempts.
- Optional batching for large query sets to avoid overload.
- Optional parallel evaluation for per-query fallback.

Usage

```
rb_eval_with_retries(
  queries,
  control,
  require_non_na = TRUE,
  eval_fn = rb_eval_single,
  batch_size = Inf,
  kill_excel_on_retry = TRUE,
  parallel_fallback = FALSE,
  mc.cores = max(1L, getOption("mc.cores", 1L))
)
```

Arguments

queries	Character vector of Excel or ProphetX formulas (e.g., RTD calls like "=RTD('prophetx.rtd', , 'ZCZ25')").
control	Named list from rexcelbridge_controls(). Uses: • control$excel_eval_max_attempts (default = 3) • control$excel_eval_retry_sleep (default = 0.15) • control$continuous_integration_session (skip evaluation in CI)
require_non_na	Logical(1). If TRUE, retries continue until all results are non-NA or attempts are exhausted. If FALSE, any length-correct result ends the loop.
eval_fn	Function used to evaluate queries (default = rb_eval_single). The function may return: • a data.frame with a result column (preferred), • a list with a $result vector, or • an atomic vector of results.
batch_size	Integer(1). Maximum number of queries per batch in vectorized evaluation (default = Inf for all-at-once).
kill_excel_on_retry	Logical(1). If TRUE, calls rb_kill_excel() before retrying after a failed attempt (default = TRUE).
parallel_fallback	Logical(1). If TRUE, uses parallel evaluation for per-query fallback when vectorized evaluation fails (default = FALSE).
mc.cores	Integer(1). Number of parallel workers used in fallback mode (default = getOption("mc.cores", 1L)).

Details

This function is primarily designed for backend Excel or ProphetX bridge environments where calls to Excel COM objects may intermittently fail or return incomplete results. It minimizes redundant evaluations and can safely handle large-scale evaluation workloads by chunking and retrying only the failed queries.

Value

Character vector of evaluated results with the same length as queries. Each entry corresponds to the evaluated output (or NA_character_ if evaluation failed after all retries).

See Also

Other excel bridge: rb_ensure_addin(), rb_eval_single(), rb_excel_quote(), rb_kill_excel(), rb_ready_predicate(), rb_start_excel()

rb_excel_quote *Quote a value for use in an Excel formula*

Description

Escapes embedded quotes and wraps non-numeric values in double quotes.

Usage

```
rb_excel_quote(x)
```

Arguments

x	Scalar to embed in an Excel formula.

Value

Character scalar safe to paste into a formula.

See Also

Other excel bridge: rb_ensure_addin(), rb_eval_single(), rb_eval_with_retries(), rb_kill_excel(), rb_ready_predicate(), rb_start_excel()

rb_kill_excel *Force close all Excel processes (Windows only)*

Description

Utility to terminate every running Excel instance on the system. Useful if RTD sessions or hidden COM objects are left running after errors or crashes. **Use with caution:** unsaved work in Excel will be lost.

Usage

```
rb_kill_excel(force = TRUE, verbose = FALSE)
```

Arguments

force	Logical; if TRUE, forcibly terminates without prompts (default).
verbose	Logical; print output from taskkill. Default FALSE.

Value

Invisibly returns the exit status code from taskkill (0 = success).

Note

Windows-only; errors on non-Windows platforms.

See Also

Other excel bridge: rb_ensure_addin(), rb_eval_single(), rb_eval_with_retries(), rb_excel_quote(), rb_ready_predicate(), rb_start_excel()

`rb_ready_predicate` *Build a readiness predicate*

Description

Returns a function(value) that decides whether a cell is "ready". By default, treats blanks, Excel errors, explicit NA values, and vendor "wait" tokens as NOT ready.

Usage

```
rb_ready_predicate(
  wait_tokens = c("Wait", "Loading...", "N/A"),
  treat_errors_as_ready = FALSE
)
```

Arguments

`wait_tokens`	Character vector of tokens treated as "not ready" (e.g., `c("Wait","Loading...","N/A")`). Case-insensitive.
`treat_errors_as_ready`	Logical; if TRUE, Excel errors stop the wait.

Value

A function `f(x)` -> `TRUE` if ready, `FALSE` otherwise.

See Also

Other excel bridge: rb_ensure_addin(), rb_eval_single(), rb_eval_with_retries(), rb_excel_quote(), rb_kill_excel(), rb_start_excel()

`rb_start_excel` *Start an Excel COM session (Windows)*

Description

Launch a hidden (by default) Excel Application via COM and configure display alerts and calculation mode (Automatic).

Usage

```
rb_start_excel(visible = FALSE)
```

Arguments

`visible`	Logical; show the Excel window. Default FALSE.

Value

A COM handle to the Excel Application object.

See Also

Other excel bridge: rb_ensure_addin(), rb_eval_single(), rb_eval_with_retries(), rb_excel_quote(), rb_kill_excel(), rb_ready_predicate()

`rexcelbridge_controls` *Controls for rexcelbridge session*

Description

Centralized knobs for batching and Excel/ProphetX evaluation behavior.

Usage

```
rexcelbridge_controls(
  query_limit = 500L,
  continuous_integration_session = FALSE,
  excel_eval_max_attempts = 2L,
  excel_eval_retry_sleep = 1
)
```

Arguments

`query_limit`	integer(1). Max rows per batch (used for chunking). Must be > 0. Default: `500L`.
`continuous_integration_session`	logical(1). If `TRUE`, skip Excel/ ProphetX evaluation and other side effects; used for CI runs. Default: `FALSE`.
`excel_eval_max_attempts`	integer(1). Max retry attempts when evaluating Excel RTD queries (per chunk). Must be >= 1. Default: `2L`.
`excel_eval_retry_sleep`	numeric(1). Sleep (seconds) between Excel RTD retry attempts. Must be >= 0. Default: `1`.

Value

Named list of control values (validated and sanitized).

Index

∗ **excel bridge**
rb_ensure_addin, 2
rb_eval_single, 2
rb_eval_with_retries, 3
rb_excel_quote, 5
rb_kill_excel, 5
rb_ready_predicate, 6
rb_start_excel, 6

rb_ensure_addin, 2, *3–7*
rb_eval_single, 2, *2*, *4–7*
rb_eval_with_retries, *2*, 3, *3*, *5–7*
rb_excel_quote, *2–4*, 5, *6, 7*
rb_kill_excel, *2–4*, 5, *5–7*
rb_kill_excel(), *4*
rb_ready_predicate, *2–5*, 6, *6, 7*
rb_start_excel, *2–5*, 6, *6*
rb_start_excel(), *2*
rexcelbridge_controls, 7
rexcelbridge_controls(), *4*